

EF1 2024

- ① We can set a lower α value. That shrinks the rejection region.
- ② (a) - Accuracy score obtained in the questionnaires could be the average score, or the cumulative score across all of the questionnaires or something else.
 - To obtain it from ChatGPT, we would ask it each question, record the obtained scores, and average them.
- (b)
 - Chatbot used
 - ↳ Levels: ChatGPT, Bard, Perplexity, ...
 - ↳ The most important because the goal of the experiment is to see which chatbot yields the best score.
 - Question difficulty
 - ↳ Levels: Hard, medium, easy, ...
 - ↳ Important blocking variable to make sure variance is isolated and the experiment is fair.
- ③ - Given that the data is continuous, the number of samples is large ($n \geq 30$) and we will be making pairwise comparisons between independent systems, I would use the 2-sample Z-test.
 - First, I'd look at the sample means of each chatbot's scores and select the chatbot w/ the highest $\bar{x}$ as my candidate.

We know that it's impossible to show, statistically that a non-candidate chatbot is better than the candidate as the test statistic would be negative which would make the p-value larger than 0.5

$$\bar{x}_n < \bar{x}_c \Rightarrow \frac{\bar{x}_{\text{non-candidate}} - \bar{x}_{\text{candidate}}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} < 0 \Rightarrow$$

$$\Rightarrow \text{p-value} > 0.5$$



That's why we don't need to test every combination, only the hypotheses of the form:

$$H_0: \mu_{\text{candidate}} \leq \mu_{\text{other}}$$

$$H_1: \mu_{\text{candidate}} > \mu_{\text{other}}$$

I would test the candidate against the others one by one, always selecting the non-candidate with the highest $\bar{x}$ that has not been tested yet.

If we never retain H_0 , we can say that the candidate provides the most accurate answers with 1- α confidence level.

If we retain H_0 even once, we cannot guarantee that any chatbot is the best with 1- α confidence, we might have to increase α and try again.

① No, that is static, we do not manipulate that variable in this experiment

② Random is better. Asking them in the same order might introduce hysteresis because LLMs retain memory of the previous prompts in the same context window.
Randomizing averages out this bias.
(Usually randomizing is better, the professor is probably looking for that answer)

③ Candidate: C1

$\sigma_1^2 \neq \sigma_2^2 \Rightarrow$ two-sample t -test

$H_0: \mu_{C1} \geq \mu_{C3}$
 $H_1: \mu_{C1} < \mu_{C3}$ } C1 against closest match (C3)

$$t_c = \frac{\bar{x}_{C1} - \bar{x}_{C3}}{\sqrt{\frac{s_{C1}^2}{n_{C1}} + \frac{s_{C3}^2}{n_{C3}}}} = \frac{28,95 - 30,71}{\sqrt{\frac{4,85^2}{14} + \frac{2,23^2}{18}}} \approx -1,258$$

$$df = \min(n_{C1} - 1, n_{C2} - 1) = 13$$

$$\rightarrow 0,10 < p\text{-value} < 0,15 \Rightarrow \\ \Rightarrow p\text{-value} > \alpha$$

we can only find p -value in the table and find p -value of t_{C1} because we are testing for LESS energy otherwise, p -value would be greater by 95

So we retain H_0 . Because of what I said in ②c, we can't know if one compiler consumes less energy than the rest with 90% confidence level.